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1. Title

Eight-Digit Security Lock System.

2. Introduction

2.1 Project Overview

Digital security systems are widely used to protect devices, information, restricted areas, and other resources from unauthorized access. A digital combination lock replaces a conventional mechanical key with an electronic verification process. This makes it possible to compare an entered code with a predefined code, provide a clear indication of successful access, monitor failed attempts, and automatically disable further access after repeated failures.

The present project implements an eight-bit binary security lock using digital logic components in Proteus. The circuit uses two 4-bit magnitude comparators for password verification, a D-type flip-flop for storing the unlock state, a 4-bit counter for counting unsuccessful attempts, logic gates for control and lockout detection, LEDs for status indication, and a 7-segment display for the failed-attempt counter

2.2 Problem Statement

The objective is to design and simulate an 8-bit binary combination lock that accepts a predefined password, unlocks when the correct password is entered, counts incorrect attempts, automatically locks the system after three unsuccessful attempts, activates an alarm indication, and restores the initial operating condition through a RESET input. The supplied project specification requires these functions to be demonstrated in Proteus.

2.3 Objectives

- Use an 8-bit binary password for access control.
- Use Group 05 and its 8-bit two's-complement value as the basis of the selected password; the supplied Proteus design uses 11111011 as the preset password.
- Turn ON a green LED when the correct password is entered.
- Count and display the number of incorrect attempts.
- Automatically lock the system after three incorrect attempts and turn ON the red alarm LED.
- Provide a RESET facility to restore the system after either successful unlocking or lockout.
- Demonstrate the complete operation through Proteus simulation.

3. System Specifications and Theoretical Background

3.1 Password Logic

The assigned group number is 5. In 8-bit binary representation, decimal 5 is 00000101. The two's complement of 00000101 is obtained by inverting the bits and adding one:

00000101 → 11111010 (one's complement) → 11111011 (two's complement).

Therefore, the 8-bit two's-complement value associated with Group 05 is 11111011. The supplied Proteus screenshot shows the fixed password set to 11111011, and this value is treated as the preset password for the report.

Item	Value
Group number	5
Group number in 8-bit binary	00000101
One's complement	11111010
8-bit two's complement	11111011
Preset password used in Proteus	11111011

3.2 Comparator Theory

The 74LS85 is a 4-bit magnitude comparator. It compares two four-bit binary words, A and B, and provides three mutually exclusive relationship outputs: A<B, A=B, and A>B. For a security lock, the equality output is the important signal because the circuit must recognize whether the entered bits are identical to the stored bits.

Two 74LS85 devices are used so that the complete 8-bit password can be compared as two four-bit groups. The equality cascade input is configured for equality comparison, while the two comparator equality outputs are combined by an AND gate. Consequently, the final MATCH signal becomes HIGH only when both four-bit sections are equal.

3.3 D Flip-Flop and State Memory

The 4013 dual D-type flip-flop is used as a state-memory element. A D flip-flop transfers the logic value present at D to Q on the active clock edge. In this project, the comparison result is sampled when ENTER is activated. When the comparison result represents a valid password, the flip-flop stores the unlock state and drives the green LED.

Using a flip-flop is important because the password-match signal is combinational and may change whenever the input switches change. The flip-flop therefore provides a stored output state instead of requiring the user to hold the password continuously.

3.4 Binary Counter and Attempt Display

The 74161 is a 4-bit synchronous binary counter. In the present design, the counter receives a clock/event associated with an unsuccessful ENTER operation. Its outputs represent the number of failed attempts in binary form. These outputs are also routed to the 7-segment display and to the lockout-detection logic.

The count sequence relevant to the project is 0000 → 0001 → 0010 → 0011, corresponding to 0, 1, 2, and 3 failed attempts. When the third failed attempt is reached, the lockout logic activates the red LED and prevents further access processing until RESET is applied.

3.5 Logic Gates and Signal Conditioning

AND gates are used to combine conditions that must be simultaneously true, such as the equality outputs from the two comparator sections and the logic used to detect the lockout state. NOT gates are used where an active-low or complementary control signal is required. A 74HC14 Schmitt-trigger inverter is used in the control path to provide a clean logic transition from the ENTER-related signal. Pull-down arrangements shown in the Proteus schematic establish defined LOW logic states when the switches are not activated.

4. System Design and Methodology

4.1 Design Flowchart

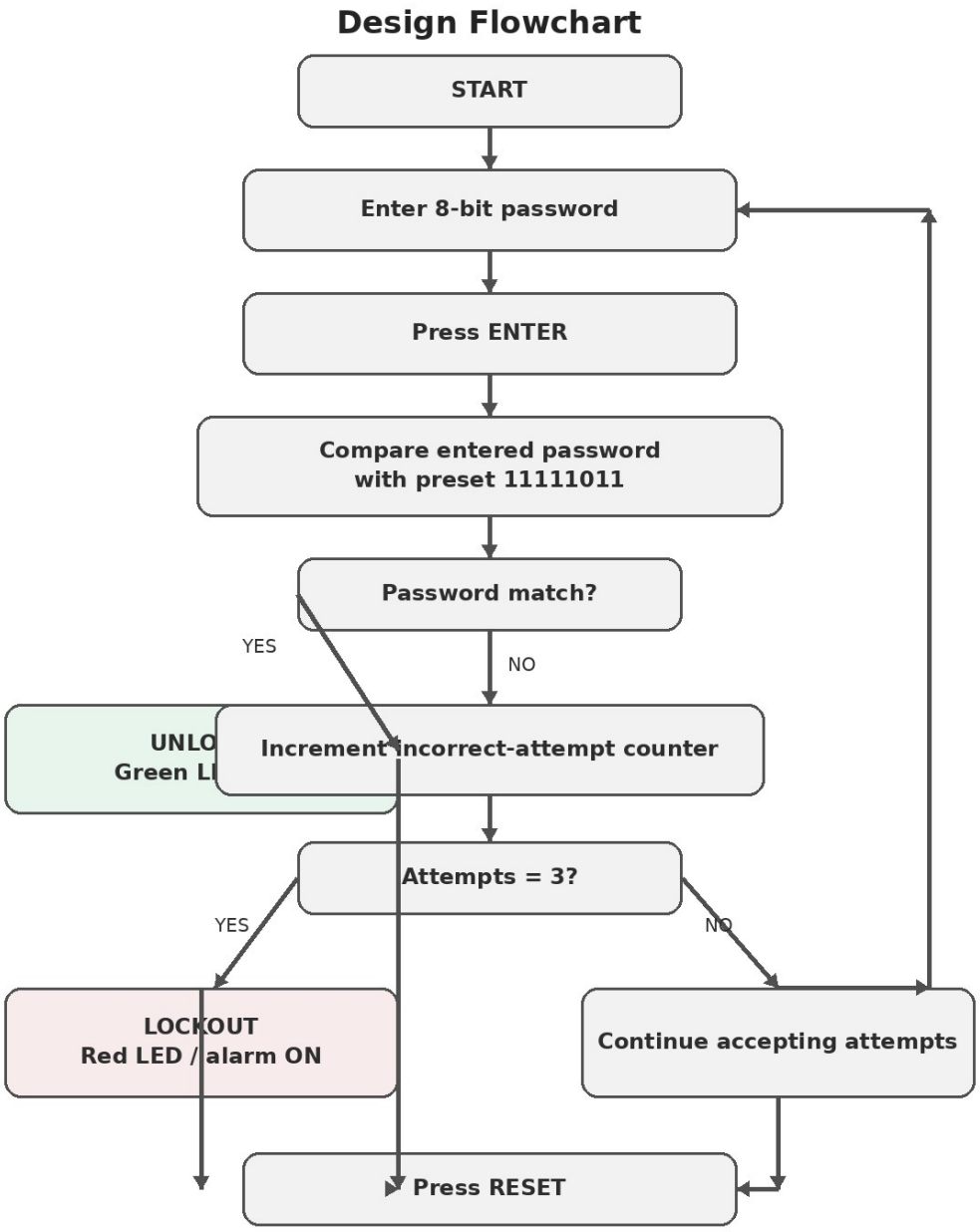


Fig-01: Project design flowchart

The flowchart represents the complete operating sequence. The user first sets an 8-bit trial password and presses ENTER. The two 74LS85 comparators determine whether the entered value is equal to the preset password. A correct match produces the unlock indication. An incorrect match increments the attempt counter. If the count reaches three, the lockout path is activated; otherwise, the system remains available for another attempt. RESET returns the system to its initial state.

4.2 Block Diagram

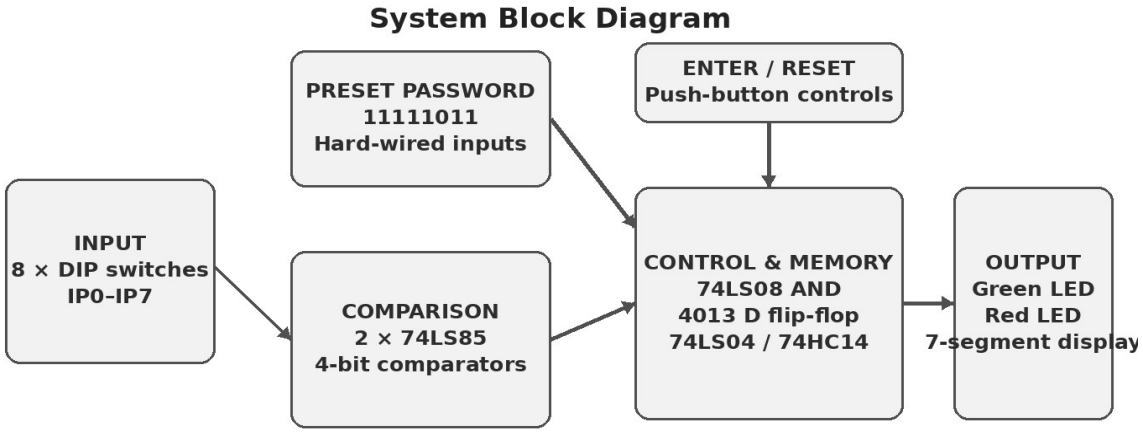


Fig-02: Functional block diagram of the security lock

The block diagram separates the design into input, password storage, comparison, control/memory, and output sections. This organization reflects the actual Proteus schematic and makes it easier to trace the signal path from the user inputs to the final indicators.

4.3 Detailed Design Process

4.3.1 Password Input and Preset Logic

The Proteus circuit uses two sets of digital logic inputs. One set represents the fixed password and the other represents the password entered by the user. The fixed-password switches are hard-wired to the selected logic pattern, while the enter-password switches can be changed during simulation. For this group, the preset value shown in the supplied schematic is 11111011.

This arrangement is simple to implement in a discrete digital circuit because the password does not require a separate memory IC or software routine. Changing the password requires changing the corresponding fixed logic states.

4.3.2 Password Comparison Stage

The eight input bits are divided into two four-bit groups. The first group is connected to U1 (74LS85), while the second group is connected to U2 (74LS85). The comparator cascade inputs are configured so that the equality relationship can be evaluated. Each comparator produces an A=B signal when its four input bits match.

The equality outputs of U1 and U2 are combined by U4, an AND gate. Therefore:

MATCH = EQ_upper × EQ_lower

where EQ_upper and EQ_lower represent the equality outputs of the two 4-bit comparator sections. MATCH = 1 only when all eight entered bits agree with the preset password.

4.3.3 Unlock Control

The MATCH signal is connected to the D input of U3:A (4013). ENTER provides the clocking event that samples the comparison result. When the user enters 11111011 and presses ENTER, both comparators

report equality, U4 produces HIGH, and the 4013 stores the HIGH state at Q. The Q output drives the green LED, proving that the system has entered the unlocked state.

For an incorrect password, MATCH remains LOW. The green LED therefore does not indicate a successful unlock, and the failed-attempt control path becomes active.

4.3.4 Incorrect Attempt Counter and Lockout

The incorrect-attempt section uses U5 (74161). The counter is advanced for an unsuccessful password-entry event. Its Q0–Q3 outputs provide the binary count and are routed toward the 7-segment display. The displayed values observed in the supplied simulation evidence are 1 after the first unsuccessful attempt, 2 after the second, and 3 when the lockout condition is reached.

The lockout network monitors the counter state and generates the control signal required to activate the red LED. At count 3 (0011), the lock condition is asserted. The associated gating and inversion stages disable the normal ENTER processing path, so additional ENTER activity does not continue increasing the count or granting access. The display remains at the lockout count until RESET is pressed.

4.3.5 Reset Logic

RESET is provided as a separate user control so that the system can recover from both successful unlock and three-attempt lockout. The reset path clears the counter and resets the stored control state. Consequently, the 7-segment display returns to 0 and the status LEDs return to their initial OFF conditions.

5. Implementation and Simulation

5.1 Proteus Simulation Setup

The circuit was implemented in Proteus using digital logic components corresponding to the functional blocks described above. The supplied schematic contains two 74LS85 comparators (U1 and U2), a 4013 flip-flop (U3:A), AND/NOT gates in the control path, a 74161 counter (U5), a 74HC14 inverter (U9:A), digital password inputs, ENTER and RESET controls, a 7-segment display, and green/red LEDs. These components and their interconnections are directly visible in the supplied Proteus screenshots.

No independent periodic clock generator is required for the demonstrated operation. ENTER acts as the user-triggered event that initiates password verification and the corresponding counter/state update. RESET is operated separately to return the circuit to the initial condition.

The reference report supplied with the project uses the same overall organization—system specification, theory, design flowchart, block diagram, detailed methodology, simulation results, walkthrough, discussion, and conclusion—while the present report follows the actual Group 05 Proteus schematic and password shown in the submitted project material.

5.2 Simulation Results

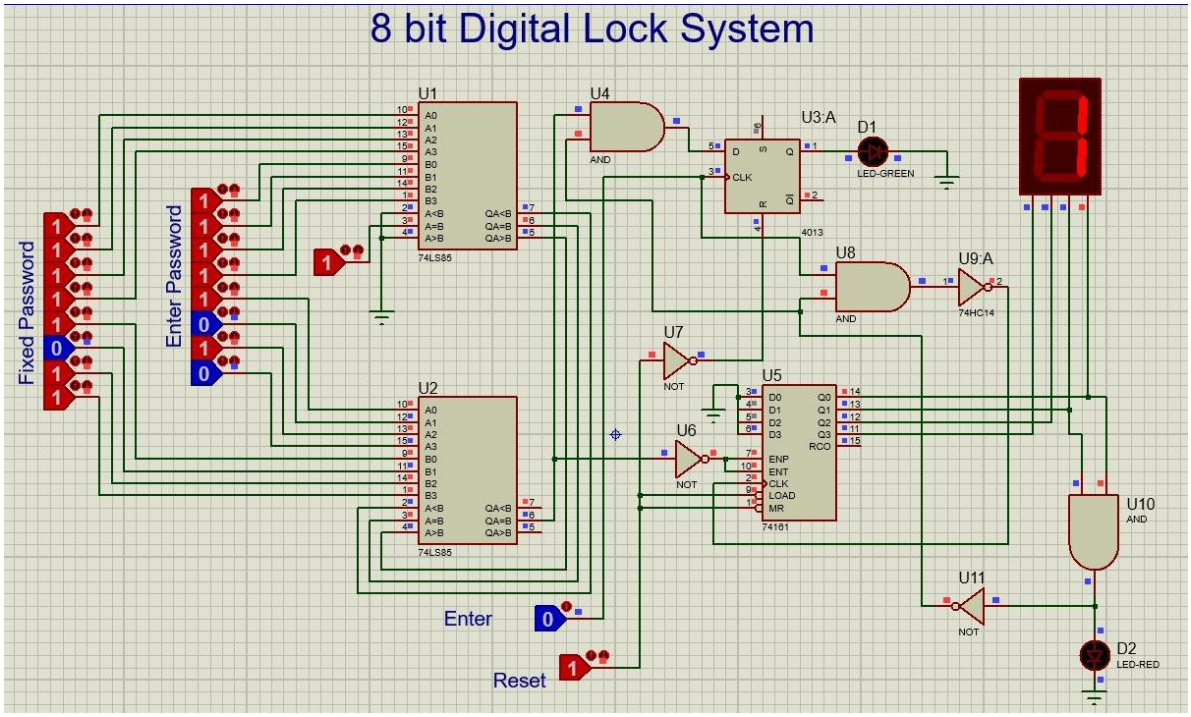


Fig-03: Proteus simulation showing one incorrect attempt; the 7-segment display indicates 1.

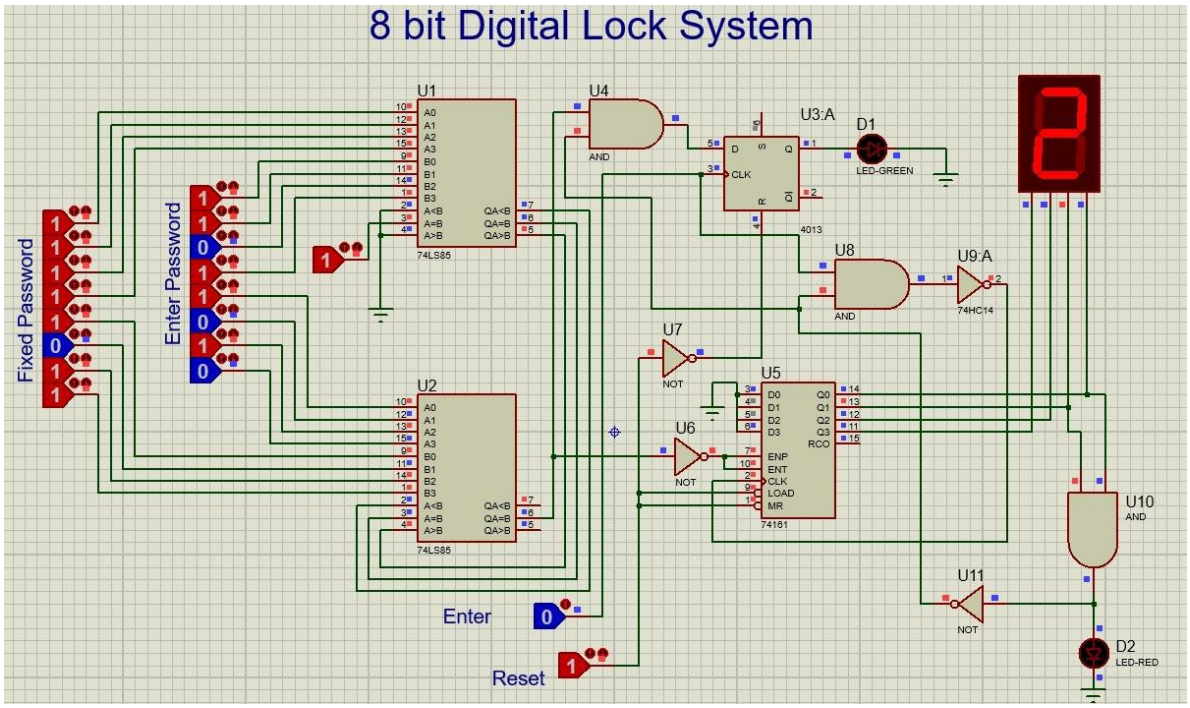


Fig-04: Proteus simulation showing two incorrect attempts; the 7-segment display indicates 2.

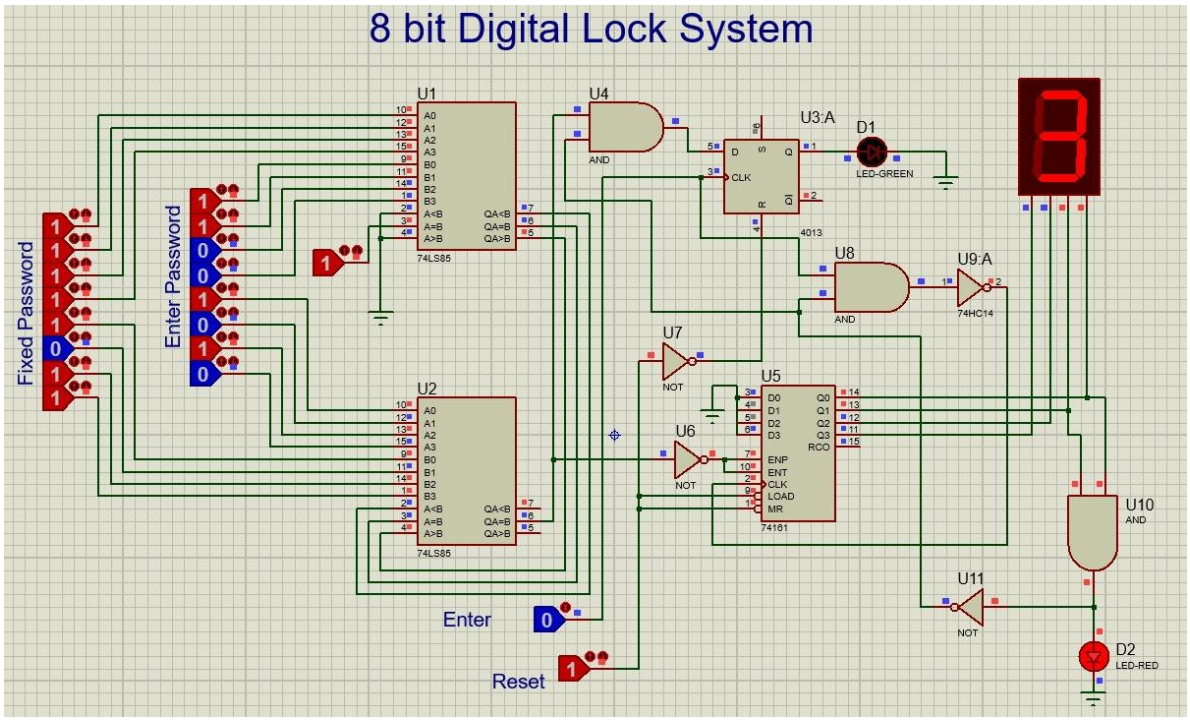


Fig-05: Proteus simulation showing the third incorrect attempt and the red alarm/lockout indication.

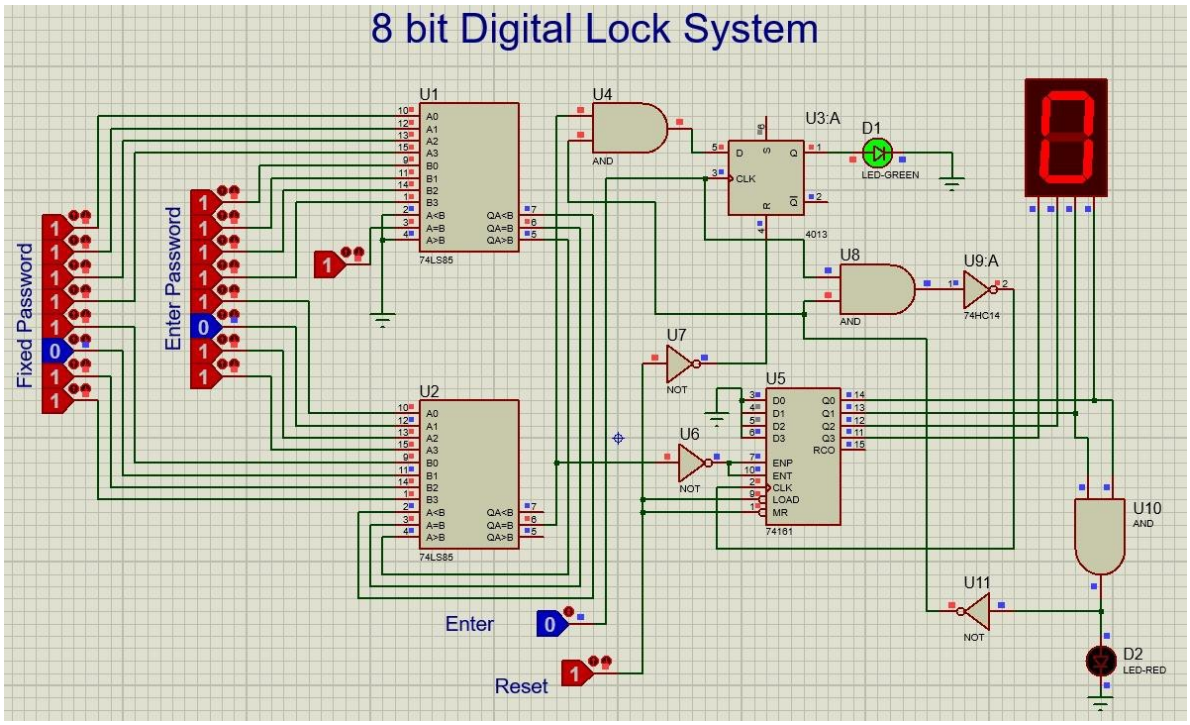


Fig-06: Proteus simulation showing the correct password condition; the green unlock LED is ON and the counter is reset to 0.

5.3 Simulation Walkthrough

5.3.1 Power-On / Reset State

At the initial condition, the counter is at zero and the stored control state is cleared. The 7-segment display therefore represents 0, while neither the green unlock indication nor the red lockout indication is active. The preset password remains fixed at 11111011.

5.3.2 User Enters an Incorrect Password

The user changes the ENTER PASSWORD switches to a value different from 11111011. The 74LS85 comparators immediately evaluate the relationship between the entered and fixed values. Since at least one bit differs, the equality output of one or both comparator sections becomes LOW. Thus, the final MATCH signal is LOW.

5.3.3 First Incorrect Attempt

When ENTER is pressed for the first wrong value, the attempt counter advances from 0 to 1. The corresponding 7-segment display shows 1. The green LED remains OFF because the password did not match. This behavior is demonstrated by the first supplied Proteus result image.

5.3.4 Second Incorrect Attempt

A second wrong password followed by ENTER increments the counter to 2. The 7-segment display changes to 2, while the system remains available for another attempt. The second supplied simulation image provides direct evidence of this state.

5.3.5 Third Incorrect Attempt and Automatic Lockout

When a third incorrect password is submitted, the counter reaches 3 (0011). The lockout detection logic responds to this state and activates the red LED. Further ENTER operations are ignored while the system is locked. The third-attempt screenshot supplied with the project shows the 7-segment display at 3 together with the red lockout indication.

5.3.6 Correct Password and Unlock

When the entered password is set to 11111011 and ENTER is pressed, both 74LS85 comparators indicate equality. U4 therefore produces a HIGH MATCH signal, which is stored by U3:A. The green LED turns ON, indicating successful access. The supplied Proteus image shows this condition together with a zero attempt count.

5.3.7 RESET Operation

RESET clears the sequential state of the system. The counter returns to zero, the 4013 stored state is cleared, the 7-segment display returns to 0, and the status indicators return to their initial state. Thus, the same circuit can be reused for a new sequence of password attempts without restarting the simulation.

6. Discussion

The implemented circuit demonstrates how combinational and sequential digital logic can be combined to form a practical security application. The comparator section is combinational: its output changes according to the relationship between the entered password and the fixed password. The 4013 and 74161 provide the required memory and counting functions, respectively. This separation is useful because the password itself can be evaluated immediately, while the result of an ENTER event and the history of previous failed attempts must be retained.

The use of two 74LS85 devices makes the eight-bit comparison straightforward. Each comparator handles four bits, and the equality outputs are combined to obtain an eight-bit match condition. This modular approach is easy to understand and troubleshoot because a mismatch can be traced to one of the two four-bit sections.

The counter and seven-segment display provide direct user feedback. Instead of merely indicating that access has failed, the circuit shows how many unsuccessful attempts have occurred. This makes the operation observable in simulation and satisfies the requirement to count incorrect attempts. The third attempt is treated differently: the lockout logic changes the system from an ordinary failed-attempt state to a protected state in which further ENTER actions are ignored.

Signal conditioning is also important in a switch-based digital circuit. Floating inputs can cause undefined logic states in simulation, while mechanical switches in practical hardware may produce bounce. The supplied design uses defined logic levels and inverter stages in the control path to reduce unwanted behavior. The reference report similarly emphasizes the importance of defined switch states, signal conditioning, and the practical use of current-limiting resistors with LEDs.

A limitation of the discrete-IC approach is that it requires several individual logic devices and interconnections. A microcontroller-based implementation could be smaller and easier to reprogram, but the discrete design is particularly suitable for an EEE283 Digital Logic Design project because the internal operation of comparators, counters, flip-flops, gates, and indicators can be observed directly.

7. Result and Interpretation

Test condition	Expected/observed result	Interpretation
Initial / RESET	7-segment = 0; LEDs OFF	System is ready for a new attempt.
Wrong password, attempt 1	7-segment = 1	First unsuccessful attempt is counted.
Wrong password, attempt 2	7-segment = 2	Second unsuccessful attempt is counted.
Wrong password, attempt 3	7-segment = 3; red LED ON	Lockout condition is activated.
Further ENTER after lockout	No further access processing	System remains locked until RESET.
Correct password 11111011	Green LED ON	Password match is confirmed and system unlocks.
RESET after unlock/lockout	Counter returns to 0; stored state clears	System is restored for a fresh operation.

The supplied simulation screenshots provide evidence for the key operating states: one failed attempt, two failed attempts, three-attempt lockout, and successful password entry. Together, these results demonstrate that the implemented circuit addresses the main functional requirements of the project.

8. Conclusion

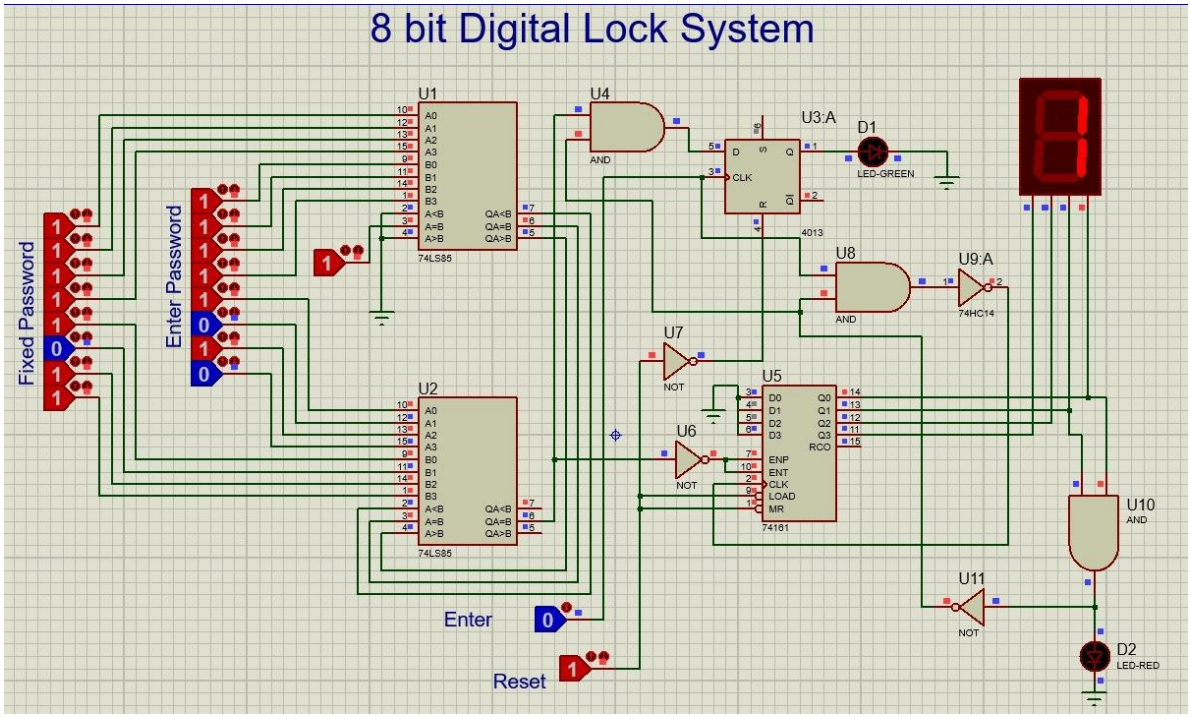
The Eight-Digit Security Lock System was successfully represented and analyzed as a digital logic design in Proteus. The design uses two 74LS85 comparators to perform the eight-bit password comparison, a 4013 D-type flip-flop to retain the unlock state, a 74161 counter to count unsuccessful attempts, logic gates and inverter stages for control and lockout, a 7-segment display for attempt indication, and green/red LEDs for system status.

For Group 05, the 8-bit two's-complement value used as the preset password in the supplied Proteus design is 11111011. The simulation evidence demonstrates the progression from zero to one and two failed attempts, automatic lockout at the third failed attempt, and successful unlocking when the correct password is entered. RESET restores the system to its initial state. Therefore, the design satisfies the required security-lock functionality and demonstrates the application of core digital logic principles in a practical system.

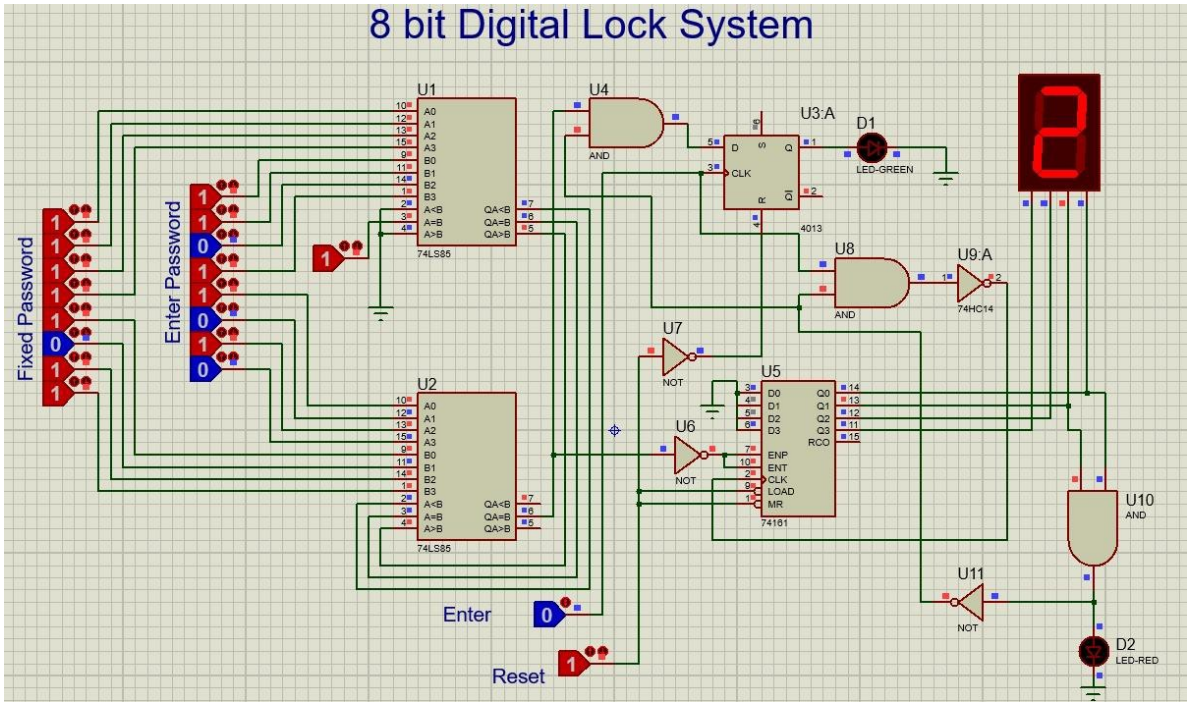
Appendix

A. Supplied Proteus schematic and output evidence

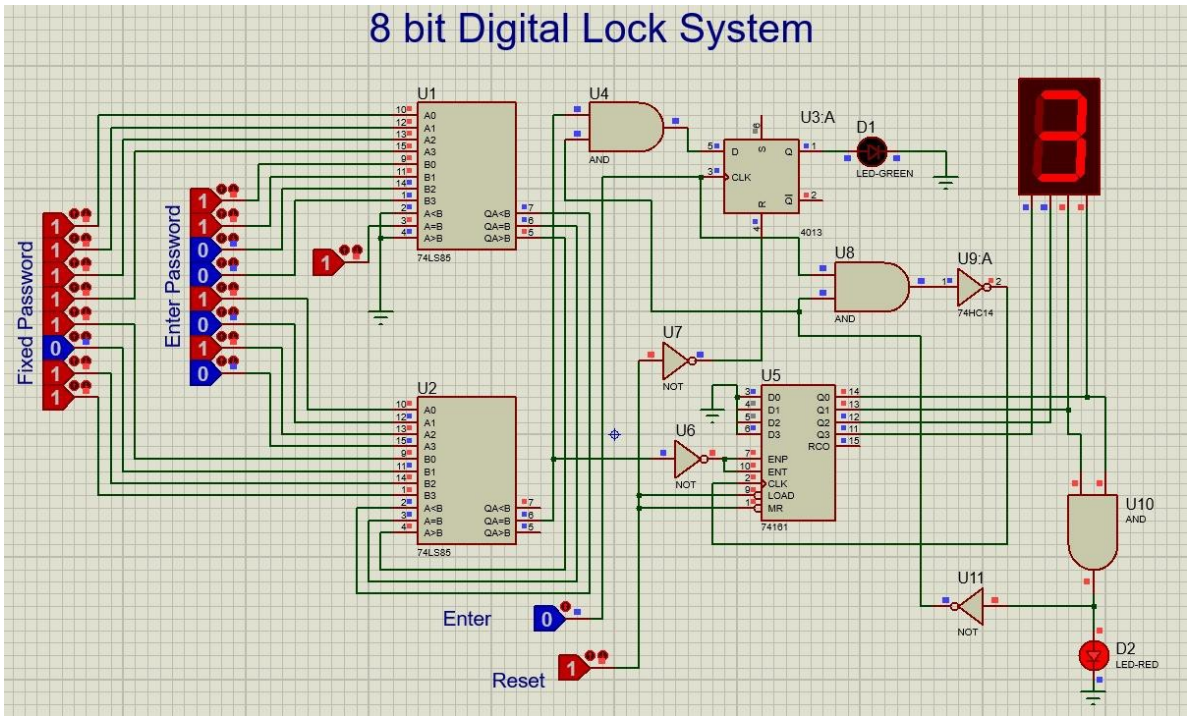
The following figures are reproduced from the Proteus screenshots supplied with the project material and are included to document the actual circuit connections and observed outputs.



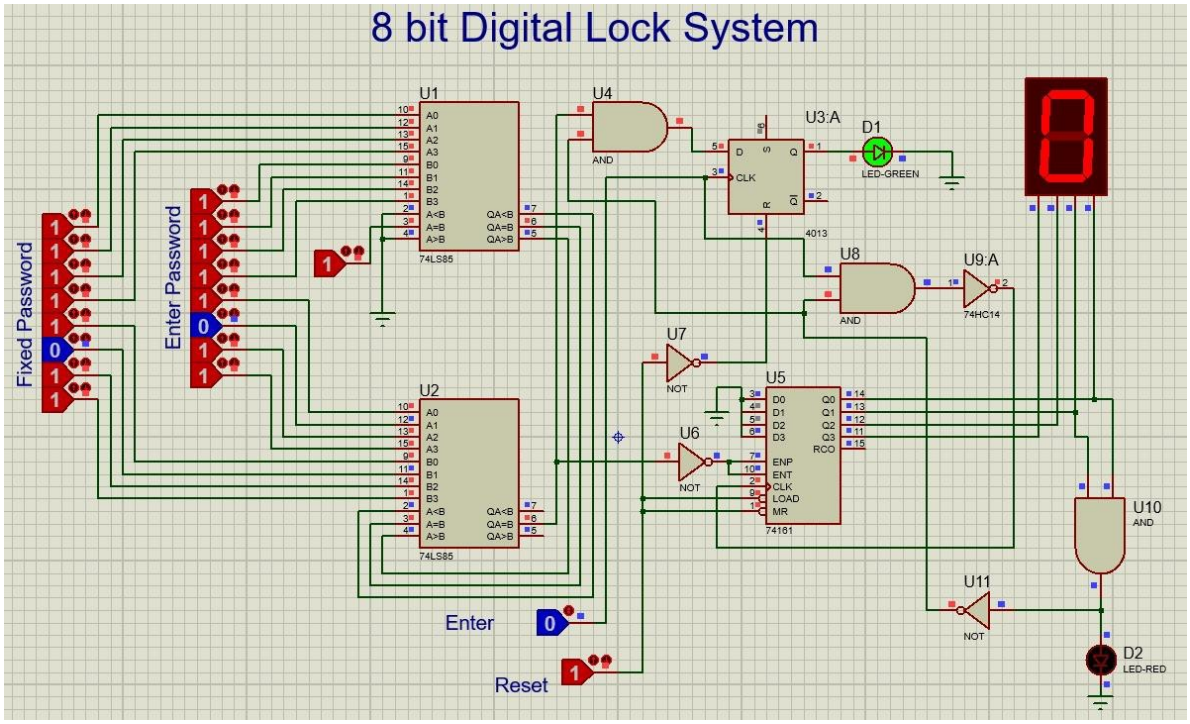
Appendix Fig-A1: Supplied Proteus schematic/output state for the first incorrect attempt.



Appendix Fig-A2: Supplied Proteus schematic/output state for the second incorrect attempt.



Appendix Fig-A3: Supplied Proteus schematic/output state for the third incorrect attempt and lockout.



Appendix Fig-A4: Supplied Proteus schematic/output state for the correct password and unlock.